

A Search-to-Decision Reduction for Continuous LWE

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Abstract

We give a search-to-decision reduction for the continuous version of the LWE problem, CLWE (Bruna, Regev, Song and Tang, STOC 2021). In 2022, Gupte, Vafa, and Vaikuntanathan showed a classical reduction from CLWE to LWE and discrete-CLWE (secret from a discrete set) to LWE. However, since the reverse reduction of CLWE to LWE was only for discrete-CLWE, there is no search-to-decision reduction for general CLWE.

We now give a relatively simple algorithm, that approximates the secret vector to within a small error. Showing the decision oracle to be powerful enough to solve the search version of the CLWE problem.

1 Introduction

In 2022, Gupte et al. [GVV22] gave a classical reduction from LWE to CLWE and discrete-CLWE to LWE. Introduced by Bruna et al. [Bru+21], the CLWE distribution is the continuous analogue of the LWE distribution. The samples from the LWE distribution is defined to be $(\mathbf{a}, b := \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q)$ where $\mathbf{a}, \mathbf{s} \sim \mathbb{Z}_q^n$ and $e \sim \chi$ where χ is some discrete distribution. The CLWE distribution is continuous and therefore $\mathbf{a}$ is a real vector possibly from a multivariate Gaussian, $\mathbf{s}$ is a unit vector from the $(n - 1)$ -dimensional sphere, e is a uniformly random sample generated from a continuous distribution, and b is modded 1.

The classical reduction from LWE to CLWE by [GVV22] is from fixed-norm LWE to CLWE. Given samples $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle (\bmod q)) \in \mathbb{Z}_q^{n+1}$, they transformed the samples to follow the CLWE distribution. They first turned

the errors in b to continuous Gaussians by adding a small continuous Gaussian to b . Next, they add a continuous Gaussian noise to $\mathbf{a}$ to turn from discrete to continuously uniform over the torus. Lastly, turn uniform $\mathbf{a}$ into Gaussian samples, scale down the secret, and randomly rotate it to ensure its a uniformly random unit vector.

The reverse reduction however, is only from discrete-CLWE to LWE. This means the CLWE samples are generated from a secret vector in a discrete set. There is no full reduction from general CLWE to LWE. Therefore, general CLWE does not currently have a search-to-decision reduction. The difficulty of the search-to-decision reduction for general CLWE lies with the secret vector being a real vector and hence, there are infinitely many guesses for each coordinate, unlike the search-to-decision reduction for LWE [MP11], where there are only $\mathbb{Z}_q$ numbers to guess.

We give a simple search-to-decision reduction for general CLWE. The process involves a simple algorithm that guesses each coordinate in steps from $-\gamma$ to γ where if the steps are small enough, the guess will be off only by a small amount.

2 Preliminaries

Unless otherwise specified, all vectors are row vectors. We let $\mathbf{u}_{n,i}$ denote the n -dimensional vector with 0 as every entry, except the i -th entry, which is 1 (i.e. a standard basis vector). For a vector $\mathbf{x}$ in $\mathbb{R}^n$ we define the l_p norm to be $\|\mathbf{x}\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ and for $p = \infty$ we define the infinite norm of $\mathbf{x}$ as $\|\mathbf{x}\|_\infty = \max_{i \in [n]} |x_i|$.

Let $I_{n \times n}$ be the $n \times n$ identity matrix. For $n \in \mathbb{N}$ we let S^{n-1} be the $(n-1)$ unit sphere or equivalently all unit vectors in $\mathbb{R}^n$. The symbol $\mathbb{T}_q$ is the torus scaled up by q where $\mathbb{T}_q = \mathbb{R}/q\mathbb{Z} = [0, q)$ and let $\mathbb{T} := \mathbb{T}_1$ be the standard torus.

Definition 2.1 (Delta function). *We define our delta function to be*

$$\delta(\alpha) = c \cdot 10^{-\alpha}$$

for any $c > 0 \in \mathbb{R}$.

Definition 2.2 ($\delta(\alpha)$ -close). *For two numbers a and b in $\mathbb{R}$, we say they are $\delta(\alpha)$ -close if*

$$|a - b| \leq \delta(\alpha)$$

Furthermore, for two vectors $\mathbf{w}$ and $\mathbf{w}'$ in $\mathbb{R}^n$, we say the two vectors are $\delta(\alpha)$ -close if

$$\|\mathbf{w} - \mathbf{w}'\|_\infty \leq \delta(\alpha)$$

Finally, for two distributions $\mathcal{D}_1$ and $\mathcal{D}_2$, we say the two distributions are $\delta(\alpha)$ -close if

$$\Delta(\mathcal{D}_1, \mathcal{D}_2) \leq \delta(\alpha)$$

2.1 Statistics

We let $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ be the multivariate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix Σ with pdf proportional to $e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$.

For arbitrary $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{c} \in \mathbb{R}^n$, we define the Gaussian function as $\rho_{s,\mathbf{c}}(\mathbf{x}) = \exp(-\pi \|(\mathbf{x} - \mathbf{c})/s\|^2)$ and we omit $\mathbf{c}$ when $\mathbf{c} = \mathbf{0}$.

Definition 2.3 (Distinguisher's advantage). *We say a distinguisher $\mathcal{A}$ has advantage ϵ in distinguishing two distributions $\mathcal{D}_1, \mathcal{D}_2$ as*

$$\left| \Pr_{x \sim \mathcal{D}_1} [\mathcal{A}(x) = 1] - \Pr_{x \sim \mathcal{D}_2} [\mathcal{A}(x) = 1] \right| = \epsilon$$

over any internal randomness of $\mathcal{A}$.

Definition 2.4 (Statistical distance). *For two distributions $\mathcal{D}_1$ and $\mathcal{D}_2$ over $\mathbb{R}^n$ with density functions ϕ_1 and ϕ_2 , respectively, we define the statistical distance between them as*

$$\Delta(\mathcal{D}_1, \mathcal{D}_2) = \frac{1}{2} \int_{\mathbb{R}^n} |\phi_1(\mathbf{x}) - \phi_2(\mathbf{x})| d\mathbf{x}.$$

Furthermore, for two random variables $X \sim \mathcal{D}_1$ and $Y \sim \mathcal{D}_2$, we define $\Delta(X, Y)$ to be equal to $\Delta(\mathcal{D}_1, \mathcal{D}_2)$.

Definition 2.5 (Convolution of probability distributions). *The probability distribution function (pdf) of a variable $Z = X + Y$ where X and Y are independent is defined to be*

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

where f_X and f_Y are the pdf of X and Y respectively.

Definition 2.6 (Kullback-Leibler divergence for univariate normal distributions ([BA11])). *For two univariate normal distributions p and q , the KL divergence between them is*

$$D_{KL}(p||q) = \ln \frac{\sigma_1}{\sigma_0} + \frac{\sigma_0^2 + (\mu_0 - \mu_1)^2}{2\sigma_1^2} - \frac{1}{2}$$

Definition 2.7 (Pinsker’s inequality). *The statistical distance between two probability distributions P and Q is bounded by*

$$\Delta(P, Q) \leq \sqrt{\frac{1}{2} D_{KL}(P||Q)}$$

Definition 2.8 (PDF of a normal distribution modulo 1 ([MJ00])). *The probability density function of a normal distribution modulo 1, where μ is the mean and σ is the standard deviation, is*

$$f_{WN}(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \sum_{k=-\infty}^{\infty} \exp\left[\frac{-(x - \mu + k)^2}{2\sigma^2}\right] = \sum_{n=-\infty}^{\infty} \phi(2\pi n) e^{i2\pi nx}$$

where $\phi(t) = e^{it\mu - \frac{1}{2}\sigma^2 t^2}$.

2.2 Learning with Errors

Definition 2.9 (CLWE distribution [GVV22]). *Let $n, m \in \mathbb{N}$, $\gamma, \beta \in \mathbb{R}$, and let $\mathcal{A}$ be a distribution over $\mathbb{R}^n$ and $\mathcal{S}$ be a distribution over S^{n-1} . Let $\text{CLWE}(m, \mathcal{A}, \mathcal{S}, \gamma, \beta)$ be the distribution given by sampling $\mathbf{a}_1, \dots, \mathbf{a}_m \sim \mathcal{A}$, $\mathbf{s} \sim \mathcal{S}$, $\mathbf{e}_1, \dots, \mathbf{e}_m \sim D_\beta$ and outputting $(\mathbf{a}_i, \gamma \cdot \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \pmod{1})$ for all $i \in [m]$. Explicitly, for one sample, the density at $(\mathbf{y}, z) \in \mathbb{R}^n \times [0, 1)$ is proportional to*

$$\mathcal{A}(\mathbf{y}) \cdot \sum_{k \in \mathbb{Z}} \rho_\beta(z + k - \gamma \cdot \langle \mathbf{y}, \mathbf{s} \rangle)$$

for fixed secret $\mathbf{s} \sim \mathcal{S}$. Unless otherwise specified, we always let $\mathcal{A}$ be the multivariate normal distribution $\mathcal{N}(\mathbf{0}, I_{n \times n})$ and $\mathcal{S}$ be the unit sphere S^{n-1} .

Definition 2.10. *The search version of the CLWE problem, denoted $\text{S-CLWE}_{\gamma, \beta}^m$, is given samples from the CLWE distribution, find $\mathbf{w}'$ such that $\mathbf{w}'$ is $\delta(\alpha)$ -close to the secret vector $\mathbf{w}$ times γ .*

Definition 2.11. *The decision version of the CLWE problem, denoted $\text{D-CLWE}_{\gamma, \beta}^m$, is to distinguish samples from the CLWE distribution and samples from the null distribution.*

3 Search-to-Decision Reduction

Theorem 3.1. *Suppose there exists a decision oracle $\mathcal{O}$ that can distinguish samples between $\text{CLWE}(m, \mathcal{A}, \mathcal{S}, \gamma, \beta)$ and $D_1^{n \times m} \times U(\mathbb{T}^m)$ with non-negligible advantage ϵ , then there exists a polynomial time reduction from $\text{S-CLWE}_{\gamma, \beta}^m$ to $\text{D-CLWE}_{\gamma, \beta}^m$.*

Proof: By Lemma 3.2, we have a polynomial time algorithm which given as input the decision oracle $\mathcal{O}$ and samples $(\mathbf{a}, b)$, returns $\mathbf{w}'$ in polynomial time. Then, by Lemma 3.3 the secret vector $\mathbf{w}'$ returned is $\delta(\alpha)$ -close to the real secret vector $\gamma \cdot \mathbf{w}$. Finally, by Lemma 3.4, the secret vector found can be used to generate CLWE samples, encrypt, and decrypt as the original CLWE distribution. ■

Lemma 3.1. *The statistical distance between the samples $(\mathbf{a}, b)$ from $\text{CLWE}(m, \mathcal{A}, \mathcal{S}, \gamma, \beta)$ and $(\mathbf{a}', b')$ where $r = -a_i + x$ and $x \sim \mathcal{N}(0, 1)$, is $\delta(\alpha)$ -close if g_i (guess of coordinate i) is $\delta(\alpha)$ -close to $\gamma \cdot w_i$.*

Proof:

$$\begin{aligned}
b' &= b + r \cdot g_i = \gamma \cdot \langle \mathbf{a}, \mathbf{w} \rangle + e + r \cdot g_i \\
&= (a_1 \cdot w_1 \cdot \gamma) + \dots + (a_i \cdot w_i \cdot \gamma + r \cdot g_i) + \dots + (a_n \cdot w_n \cdot \gamma) + e \\
&= (a_1 \cdot w_1 \cdot \gamma) + \dots + (a_i \cdot w_i \cdot \gamma + r \cdot w_i \cdot \gamma - r \cdot w_i \cdot \gamma + r \cdot g_i) + \dots + (a_n \cdot w_n \cdot \gamma) + e \\
&= (a_1 \cdot w_1 \cdot \gamma) + \dots + (\gamma w_i (a_i + r) + r(g_i - \gamma w_i)) + \dots + (a_n \cdot w_n \cdot \gamma) + e \\
&= (a_1 \cdot w_1 \cdot \gamma) + \dots + (\gamma w_i (a_i + r)) + \dots + (a_n \cdot w_n \cdot \gamma) + e + r(g_i - \gamma w_i) \\
&= \gamma \cdot \langle \mathbf{a}', \mathbf{w} \rangle + e + r(g_i - \gamma w_i) \\
&= \gamma \cdot \langle \mathbf{a}', \mathbf{w} \rangle + e'
\end{aligned}$$

We now show correctness. First, $a_i + r = a_i + (-a_i + x) = x$ follows $\mathcal{N}(0, 1)$ and therefore $\mathbf{a}'$ follows $\mathcal{N}(\mathbf{0}, I_{n \times n})$. We know that $|g_i - \gamma \cdot w_i| \leq \delta(\alpha)$ and therefore $r|g_i - \gamma \cdot w_i| \leq r \cdot \delta(\alpha)$. Hence, $e' = e + r(g_i - \gamma \cdot w_i) \leq e + r \cdot \delta(\alpha)$.

By Claim 3.1, the statistical distance between e and e' is $\delta(\alpha)$ -close and therefore $(\mathbf{a}', b')$ follows the CLWE distribution for a large enough α (will be chosen later). ■

Claim 3.1. *The statistical distance between the continuous Gaussian distribution $\mathcal{N}(0, \beta^2)$ and its convolution with the continuous Gaussian distribution $\mathcal{N}(0, 2)$ multiplied by $\delta(\alpha)$ is $\delta(\alpha)$ -close.*

Proof: Let $X \sim \mathcal{N}(0, \beta^2)$, $Y \sim \mathcal{N}(0, 2\delta(\alpha)^2)$, and $Z = X + Y$. Then $Z \sim \mathcal{N}(0, \beta^2 + 2\delta(\alpha)^2)$. By Definition 2.6, we know that

$$\begin{aligned}
D_{KL}(Z||X) &= \ln \left(\frac{\beta^2}{\beta^2 + 2\delta(\alpha)^2} \right)^{1/2} + \frac{\beta^2 + 2\delta(\alpha)^2 + (0 - 0)^2}{2(\beta^2)} - \frac{1}{2} \\
\text{Let } x &= \frac{2\delta(\alpha)^2}{\beta^2}
\end{aligned}$$

$$\begin{aligned}
\rightarrow \frac{1}{2}(-\ln(1+x) + (1+x) - 1) &\approx \frac{1}{2}(-x + \frac{x^2}{2} + (1+x) - 1) \\
&= \frac{x^2}{4} = \frac{4\delta(\alpha)^4}{4\beta^4} = \frac{\delta(\alpha)^4}{\beta^4}
\end{aligned}$$

Using Pinsker's inequality:

$$\Delta(Z, X) \leq \sqrt{\frac{\delta(\alpha)^4}{2\beta^4}}$$

Therefore, Z and X is $\delta(\alpha)$ -close. ■

Lemma 3.2. *There exists an efficient polynomial algorithm, where given a decision oracle $\mathcal{O}$ and samples from $\text{CLWE}(m, \mathcal{A}, \mathcal{S}, \gamma, \beta)$, guesses the secret vector $\mathbf{w}'$ $\delta(\alpha)$ -close to $\gamma \cdot \mathbf{w}$.*

Proof: We now show a polynomial time algorithm where given $\mathcal{O}$ and samples $(\mathbf{a}, b)$ guesses the secret vector $\gamma \cdot \mathbf{w}$. First, let $m, l = \text{poly}(n)$ and let $\alpha = O(\log_2 n)$. We know that $\gamma = \tilde{\Omega}(\sqrt{n})$ ([GVV22]) and in general $\gamma \approx \sqrt{n}$. Therefore we choose γ to be $c_1 \sqrt{n}$ for some constant c_1 .

For each coordinate w_i , the range of our guess is $-\gamma$ to γ in $\delta(\alpha)$ steps, i.e. our guess g_i starts from $-\gamma$ and the next guess increases by $\delta(\alpha)$ until we reach γ . For each guess g_i we generate m samples $(\mathbf{a}' = \mathbf{a} + r \cdot \mathbf{u}_{n,i}, b' = b + r \cdot g_i)$ (Lemma 3.1) where $r = -a_i + x$.

We pass the samples to the distinguisher $\mathcal{O}$ and store whether it accepts or rejects. We then repeat this for $l - 1$ more times before moving on to the next guess. After iterating through the entire range, pick the coordinate that had the most amount of accepts.

We now show correctness. First, the algorithm runs in polynomial time, and we use a polynomial number of samples. Next, since each coordinate of $\mathbf{w}$ is multiplied by γ , we let our range of guess be $[-1, 1] \cdot \gamma$.

Notice that the error added is $(-a_i + x)(g - \gamma w_i)$ where $a_i, x \sim \mathcal{N}(0, 1)$ and therefore $(-a_i + x) \sim \mathcal{N}(0, 2)$. By Claim 3.2, the statistical distance between $\mathcal{N}(0, 2) \bmod 1$ and $U[0, 1]$ is essentially negligible for all practical purposes. Hence, when our guess is incorrect by a significant amount we have essentially added a uniform random variable $U[0, 1]$ and any random variable X added with a random variable from $U[0, 1] \bmod 1$ is still $U[0, 1]$.

In Claim 3.1, we have shown that when our guess is within $\delta(\alpha)$, then the statistical distance between them is $\sqrt{\frac{\delta(\alpha)^4}{2\beta^4}}$. We now show that given

$m = n^k$ samples to the oracle, for a big enough α the oracle will accept the samples with high probability. First, the divergence of Z and X (Claim 3.1) is $D_{KL}(Z||X) = \frac{\delta(\alpha)^4}{\beta^4}$. The divergence for n^k samples is

$$D_{KL}(Z^{n^k}||X^{n^k}) \approx n^k \frac{\delta(\alpha)^4}{\beta^4}$$

Using Pinsker's inequality the statistical distance is bounded by

$$\Delta(Z^{n^k}, X^{n^k}) \leq \sqrt{n^k \frac{\delta(\alpha)^4}{2\beta^4}}$$

We need $\sqrt{n^k \frac{\delta(\alpha)^4}{2\beta^4}} \ll \epsilon$. Since α is $O(\log n)$, our delta function is $10^{-c_2 \log n}$ for some constant c_2 . This gives

$$\frac{1}{10^{4c_2 \log n}} \ll \frac{2\beta^4 \epsilon^2}{n^k}$$

Simplifying and treating $8\beta^4 \epsilon^2$ as a constant we only need

$$n^{4c_2 \log 10} > n^k \rightarrow 4c_2 \log 10 > k \rightarrow c_2 > \frac{k}{4 \log 10}$$

Our oracle $\mathcal{O}$ has base advantage ϵ , i.e.

$$\left| \Pr_{x \sim \mathcal{D}_1} [\mathcal{O}(x) = 1] - \Pr_{x \sim \mathcal{D}_2} [\mathcal{O}(x) = 1] \right| = \epsilon$$

where $\mathcal{D}_1$ is the CLWE distribution and $\mathcal{D}_2$ the null distribution. Let $\mathcal{D}_3$ be the distribution of the samples as defined above. We have shown the statistical distance between $\mathcal{D}_1$ and $\mathcal{D}_3$ given n^k samples is no more than $\sqrt{n^k \frac{\delta(\alpha)^4}{2\beta^4}} = \mu(c_2)$. Therefore, our distinguisher $\mathcal{O}$ has advantage

$$\left| \Pr_{x \sim \mathcal{D}_1} [\mathcal{O}(x) = 1] - \Pr_{x \sim \mathcal{D}_3} [\mathcal{O}(x) = 1] \right| \leq \mu(c_2)$$

in distinguishing samples from $\mathcal{D}_1$ and $\mathcal{D}_3$. Replacing $\mathcal{D}_1$ with $\mathcal{D}_3$ we get

$$\left| \Pr_{x \sim \mathcal{D}_3} [\mathcal{O}(x) = 1] - \Pr_{x \sim \mathcal{D}_2} [\mathcal{O}(x) = 1] \right| \geq \epsilon - \mu(c_2)$$

Therefore, for a big enough selection of c_2 and when our guess is close enough to the real coordinate, our oracle retains its advantage and given m samples it will accept our samples with high probability.

We repeat this l times for each guess to make sure it is correct. Since our secret is continuous, it is essentially impossible to have our guess exactly the same as the real coordinate. As our guess gets closer to the real coordinate, our samples slowly transition from uniform to CLWE. Since there is no clean cut off like LWE, we need to repeat this l times and take the guess with the most accepts, to ensure we do not get a false positive.

Therefore, the coordinate we choose will be within $\delta(\alpha) \cdot C$ for some small constant $C > 0$. ■

Claim 3.2. *The statistical distance between $\mathcal{N}(0, 2) \bmod 1$ and the uniform distribution $U[0, 1]$ is essentially negligible for all practical purposes.*

Proof: By Definition 2.8, the pdf of $\mathcal{N}(0, 2) \bmod 1$ is:

$$\begin{aligned}
& \sum_{n=-\infty}^{\infty} e^{-\frac{1}{2}2(2\pi n)^2} e^{i2\pi nx} = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2} e^{i2\pi nx} \\
& \rightarrow e^{-4\pi^2 0^2} e^{i2\pi 0x} + \sum_{n=-\infty}^{-1} e^{-4\pi^2 n^2} e^{i2\pi nx} + \sum_{n=1}^{\infty} e^{-4\pi^2 n^2} e^{i2\pi nx} \\
& \quad 1 + \sum_{n=-1}^{\infty} e^{-4\pi^2 n^2} (e^{i2\pi nx} + e^{-i2\pi nx}) \\
& \text{Euler's formula} \rightarrow 1 + \sum_{n=1}^{\infty} 2 \cos(2\pi nx) e^{-4\pi^2 n^2} \\
& \approx 1 + 2 \cos(2\pi x) e^{-4\pi^2}
\end{aligned}$$

The statistical distance between $\mathcal{N}(0, 2)$ and $U[0, 1]$ is:

$$\begin{aligned}
& \frac{1}{2} \int_0^1 \left| 1 + 2 \cos(2\pi x) e^{-4\pi^2} - 1 \right| dx \\
& = \frac{e^{-4\pi^2}}{2\pi} \frac{2}{\pi} \approx 7.25 \times 10^{-19}
\end{aligned}$$
■

Lemma 3.3. *The secret vector $\mathbf{w}'$ generated by Lemma 3.2 is $\delta(\alpha)$ -close to the real secret vector $\gamma \cdot \mathbf{w}$.*

Proof: Let $\mathbf{w}'$ be the vector returned by Lemma 3.2. We know each coordinate is off from the real coordinate times γ by no more than $\pm\delta(\alpha)$. To show that $\mathbf{w}'$ is $\delta(\alpha)$ -close to $\mathbf{w}$, we show the infinite norm of $\mathbf{w}' - \gamma \cdot \mathbf{w}$ is less than or equal to $\delta(\alpha)$. Here $\delta_i \leq \delta(\alpha)$.

$$\begin{aligned}\mathbf{w}' - \gamma \cdot \mathbf{w} &= (w'_1 - \gamma \cdot w_1, \dots, w'_n - \gamma \cdot w_n) \\ \mathbf{w}' - \gamma \cdot \mathbf{w} &= ((\gamma \cdot w_1 \pm \delta_1) - \gamma \cdot w_1, \dots, (\gamma \cdot w_n \pm \delta_n) - \gamma \cdot w_n) \\ \mathbf{w}' - \gamma \cdot \mathbf{w} &= (\pm\delta_1, \dots, \pm\delta_n) \\ &\rightarrow \|\mathbf{w}' - \gamma \cdot \mathbf{w}\|_\infty \leq \delta(\alpha)\end{aligned}$$

Therefore $\mathbf{w}'$ and $\gamma \cdot \mathbf{w}$ is $\delta(\alpha)$ -close. ■

The next Lemma will use a simple public key cryptosystem to encrypt a bit. The PKC is:

- **Private Key:** Choose $\mathbf{w} \in S^{n-1}$ uniformly at random.
- **Public Key:** The public key is $(\mathbf{a}, b = \gamma \cdot \langle \mathbf{a}, \mathbf{w} \rangle + e \mod 1)$ where $\mathbf{a} \sim \mathcal{N}(\mathbf{0}, I_{n \times n})$ and $e \sim \mathcal{N}(0, \beta^2)$.
- **Encryption:** To encrypt a bit $m \in \{0, 1\}$, add $m/2$ to b , i.e., send $(\mathbf{a}, b + \frac{m}{2} \mod 1)$.
- **Decryption:** Compute $d = b - \gamma \cdot \langle \mathbf{a}, \mathbf{w} \rangle \mod 1$. If d is closer to 0 than 0.5 output 0, else output 1.

Lemma 3.4. *The secret vector $\mathbf{w}'$ returned by Lemma 3.2, can be used to generate CLWE samples, encrypt, and decrypt as the original CLWE distribution.*

Proof: To generate samples as the original CLWE distribution, generate $\mathbf{a} \sim \mathcal{N}(\mathbf{0}, I_{n \times n})$ and $e \sim \mathcal{N}(0, \beta^2)$. Then our CLWE sample is $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{w}' \rangle + e \mod 1)$.

$$\begin{aligned}\langle \mathbf{a}, \mathbf{w}' \rangle &= a_1 \cdot w'_1 + \dots + a_n \cdot w'_n \\ &= a_1(\gamma \cdot w_1 \pm \delta_1) + \dots + a_n(\gamma \cdot w_n \pm \delta_n) \\ &= \gamma \cdot a_1 \cdot w_1 + a_1 \cdot \pm\delta_1 + \dots + \gamma \cdot a_n \cdot w_n + a_n \cdot \pm\delta_n \\ &= \gamma \cdot \langle \mathbf{a}, \mathbf{w} \rangle + (a_1 \cdot \pm\delta_1 + \dots + a_n \cdot \pm\delta_n) \\ &= \gamma \cdot \langle \mathbf{a}, \mathbf{w} \rangle + \langle \mathbf{a}, \boldsymbol{\delta} \rangle\end{aligned}$$

Our CLWE sample is $\gamma \cdot \langle \mathbf{a}, \mathbf{w} \rangle + e + \langle \mathbf{a}, \boldsymbol{\delta} \rangle \pmod{1}$ and so our error is $e + \langle \mathbf{a}, \boldsymbol{\delta} \rangle$. We know $\mathbf{a} \sim \mathcal{N}(\mathbf{0}, I_{n \times n})$ and $\|\boldsymbol{\delta}\|_p \leq \delta(\alpha)$ therefore $\langle \mathbf{a}, \boldsymbol{\delta} \rangle \sim \mathcal{N}(0, \|\langle \mathbf{a}, \boldsymbol{\delta} \rangle\|_2^2) \leq \mathcal{N}(0, n \cdot c \cdot 10^{-2\alpha})$. Then $e + \langle \mathbf{a}, \boldsymbol{\delta} \rangle \sim \mathcal{N}(0, \beta^2 + n \cdot c \cdot 10^{-2\alpha})$. For a large enough α , $\mathcal{N}(0, \beta^2 + n \cdot c \cdot 10^{-2\alpha}) \approx \mathcal{N}(0, \beta^2)$.

We now show encryption with *our* vector $\mathbf{w}'$ and decryption with the *real* vector $\mathbf{w}$. To encrypt a bit m , add $m/2$ to b in $(\mathbf{a}, b)$. For bit 0 we have $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{w}' \rangle + e + \langle \mathbf{a}, \boldsymbol{\delta} \rangle) \pmod{1}$ and for 1 we have $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{w}' \rangle + e + \langle \mathbf{a}, \boldsymbol{\delta} \rangle + \frac{m}{2}) \pmod{1}$.

Decrypting with the real $\gamma \cdot \mathbf{w}$ we have

$$d = b - \gamma \cdot \langle \mathbf{a}, \mathbf{w} \rangle = e + \langle \mathbf{a}, \boldsymbol{\delta} \rangle + \frac{m}{2} \pmod{1}$$

Let $e' = e + \langle \mathbf{a}, \boldsymbol{\delta} \rangle$ then to correctly decrypt we only need $|e'| < 0.25$.

Next, we show encryption with the *real* secret vector $\mathbf{w}$ and decryption with *our* vector $\mathbf{w}'$. Encrypting a bit m we have $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{w} \rangle + e + \frac{m}{2} \pmod{1})$ and decrypting

$$d = b - \langle \mathbf{a}, \mathbf{w}' \rangle = e - \langle \mathbf{a}, \boldsymbol{\delta} \rangle + \frac{m}{2} \pmod{1}$$

Let $e' = e - \langle \mathbf{a}, \boldsymbol{\delta} \rangle$. Then, to correctly decrypt we, again, only need $|e'| < 0.25$. ■

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